

where $\eta > 0$ is the step-rate, k is the time step, $\mathcal{N}_i$ denotes the set of neighbours of agent i , W_{ij} is the consensus weighting factor on the shared information over link (j, i) , $0 \leq \mu < 1$ is the momentum-rate, and variable $y_i(k)$ denotes the momentum term of node i at time k . *The momentum term y_i is added to improve the convergence rate and accelerate the solution.* For initialization, the agents set their initial state values as $x_i(0) = b_i$ and $y_i(0) = 0$.

3.2. Solution Subject to Sector-Bound Nonlinearity

The data transmission channels (or the communication links) between agents might be subject to nonlinear constraints. This implies that the sent data $\partial_{x_j} f_j(k)$ over a link (j, i) is delivered to agent i as $g_l(\partial_{x_j} f_j(k))$, where $g_l : \mathbb{R} \mapsto \mathbb{R}$ denotes the nonlinear mapping. Then, the decentralized solution is updated as,

$$x_i(k+1) = x_i(k) + \mu y_i(k) + \eta \sum_{j \in \mathcal{N}_i} W_{ij} (g_l(\partial_{x_j} f_j(k)) - g_l(\partial_{x_i} f_i(k))), \quad (12)$$

$$y_i(k+1) = x_i(k+1) - x_i(k), \quad (13)$$

where the weight matrix W is only weight-balanced and not necessarily stochastic.

Assumption 3. *The nonlinear function $g_l(\cdot)$ is assumed to be odd, strongly sign-preserving, and sector-bound.*

One example of such link nonlinearity is saturation or clipping. Another example is log-scale quantization [55, 56] illustrated in Fig. 2 with its formula as follows:

$$g_l(u) := \text{sgn}(u) \exp(g_u(\log(|u|))), \quad (14)$$

with $g_u(u) := \rho \left\lceil \frac{u}{\rho} \right\rceil$ (the operator $\lceil \cdot \rceil$ as rounding to the nearest integer) and $\rho > 0$ as the quantization level.